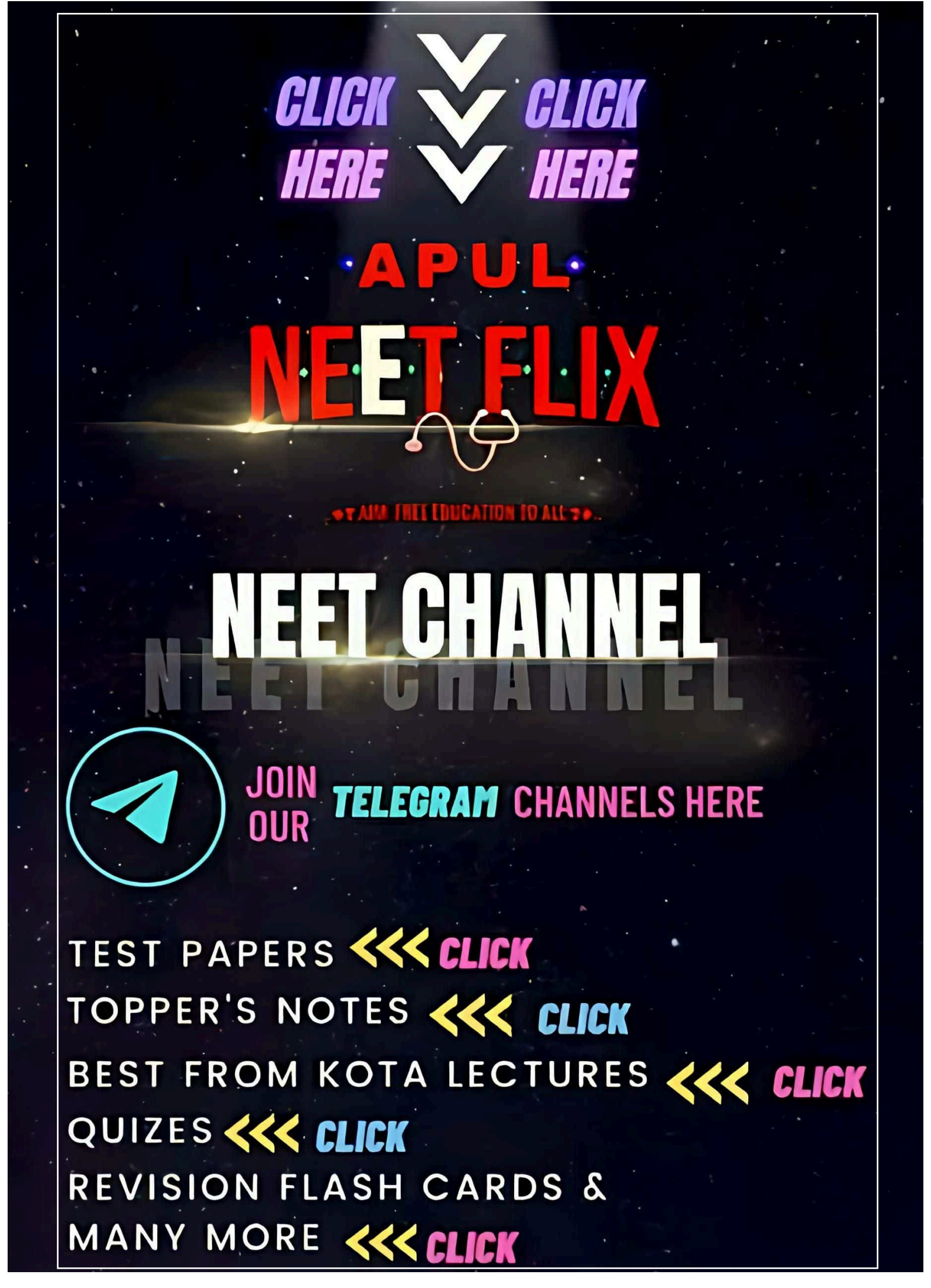

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BIOLOGY

1) Ecology at the level of organism tries to understand how different organisms are adapted to their environment in terms of not only survival but also reproduction. Thus referred as :-

(1) synecology
(2) Physiological ecology
(3) behavioural ecology
(4) developmental ecology

2) Ecology term was coined by ?

(1) E.P. Odum
(2) E.Haeckel
(3) A.G.Tansley
(4) Reiter

3) The basic unit of ecological study is :-

(1) Species
(2) Organism
(3) Community
(4) Biosphere

4) Ecosystem properties can not be detected at how many levels given below :-
Deme, Metapopulation, Biosphere, Biome, Organism, Communities, Species, Landscape, Organ system

(1) 5
(2) 6
(3) 4
(4) 3

5) **Statement–I** :- A species which is found only in a particular area is known as endemic species.
Statement–II :- *Metasequoia* is found only in valley of china.

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement I** is incorrect but **Statement II** is correct

6) **Statement–I** :- The activities of key-stone species determine the structure of the community.

Statement–II :- Fig tree in tropical forest is key-stone species.

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement I** is incorrect but **Statement II** is correct

7) Maximum abundance of photo-auto-trophs in oceans, will be found in :-

(1) Benthic zone
(2) Euphotic zone
(3) Aphotic zone
(4) Abyssial zone

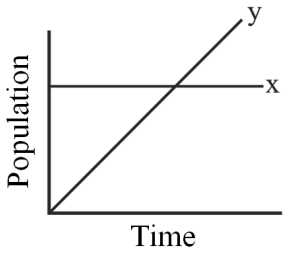
8) Relation between fig tree and wasp species :

(1) Commensalism
(2) Mutualism
(3) Parasitism
(4) Amensalism

9) The ..... are association between fungi & the roots of higher plants.

(1) Lichens
(2) *Mycorrhizae*
(3) Sexual Deceit
(4) Brood Parasitism

10) What type of interaction shows in this graph between two organism x and y :



(1) Mutualism
(2) Commensalism
(3) Amensalism
(4) Predation

11) Which of the following is an epiphyte ?

(1) Orchid
(2) Lianas
(3) Santalum

(4) Mango

12) Match the column I with column II

Column I		Column II	
(i)	*Cuscuta*	(a)	Hemi parasite
(ii)	Lianas	(b)	Holo parasite
(iii)	*Viscum*	(c)	Commensal

(1) (i)-(c), (ii)-(a), (iii)-(b)

(2) (ii)-(a), (ii)-(b), (iii)-(c)

(3) (i)-(c), (ii)-(b), (iii)-(a)

(4) (i)-(b), (ii)-(c), (iii)-(a)

13) Roots of black walnut (juglans nigra) secrete a chemical juglone which is toxic to other plants like apple, alfa alfa etc. This kind of interaction is :-

(1) Amensalism

(2) Mutualism

(3) Protocooperation

(4) Commensalism

14) Cattle or goat are never browsing on calotropis growing in abandoned field because of the presence of :-

(1) Alkaloid like Quinine

(2) Strychine

(3) Cardiac glycosides

(4) Opium/Morphine

15) Select the incorrect statement :-

(1) Carnivores appear to be more adversely affected by competition than Herbivores.

(2) Species diversity in a community depends upon size of area.

(3) The cattle egret bird and grazing cattle is an example of commensalism.

(4) Clear stratification is found in tropical rain forest.

16) Which of the following is not a favourable factor for increasing the rate of decomposition ?

(1) Oxygen richness of soil.

(2) Detritus rich in lignin

(3) Warm environment

(4) High soil moisture

17) The variation in which environmental factor shows major effect on the rate of metabolism in

organisms?

(1) Rainfall
(2) Temperature
(3) Light
(4) Water

18) Ecosystem are not exempt from the :–

(1) First law of thermodynamics
(2) Second law of thermodynamics
(3) Third law of thermodynamics
(4) Both (1) & (2)

19) Which type of productivity is used in formation of tissues in plant ?

(1) GPP
(2) GSP
(3) NPP
(4) NSP

20) Interaction of biotic and abiotic components result in a physical structures that is characteristic for each type of :-

(1) Species
(2) Population
(3) Community
(4) Ecosystem

21) Stratification is :-

(1) Identification and enumeration of plant and animal species of an ecosystem
(2) Species composition
(3) Vertical distribution of different species occupying different levels
(4) Active nature of all the components of the ecosystem as a unit

22) Choose the odd one :-

(1) Forest
(2) Grassland
(3) Desert
(4) Garden

23) Habitat includes :-
(a) Biotic component
(b) physio chemical component
(c) pathogens, parasite, predator, competitors

(d) different species

(1) Only a, b, d
(2) Only a, d
(3) Only b, c, d
(4) a, b, c, d

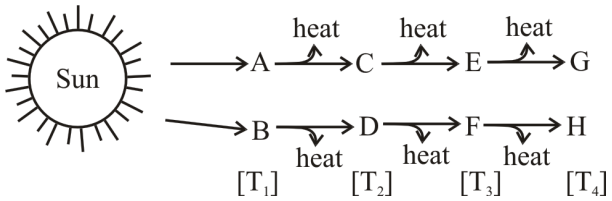
24) Who is the father of ecosystem ecology?

(1) A. G Tansley
(2) Eugene P. Odum
(3) Ram Dev Mishra
(4) Haeckel

25) Which of the following is considered as smallest structural & functional unit of nature?

(1) Organism
(2) Ecosystem
(3) Community
(4) Population

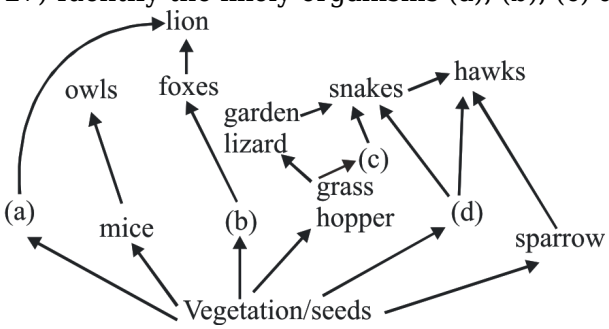
26)



How many statements are correct in relation to the given energy flow diagram.
(a) E and F are Primary carnivors
(b) C and D are herbivores
(c) G and H are top carnivores
(d) A and B are transducer

(1) One
(2) Two
(3) Three
(4) Four

27) Identify the likely organisms (a), (b), (c) and (d) in the food web shown below :



	(a)	(b)	(c)	(d)
(1)	rat	dog	tortoise	pigeon
(2)	squirrel	cat	rat	pigeon
(3)	deer	rabbit	frog	rat
(4)	dog	squirrel	bat	deer

(1) 1

(2) 2

(3) 3

(4) 4

28) Which statement is incorrect ?

(1) Trophic level represent functional level not a species as such

(2) A same species may occupy more than one trophic level in same ecosystem at same time

(3) In most ecosystem, pyramids of number, energy and biomass are upright

(4) Energy at higher trophic level is more as compared to lower levels as they are physically strong

29) With respect to energy flow which of the following is correct.

(1) Out of the total incident solar radiation less than 50% of it is PAR

(2) plants capture only 2-10% of the PAR and this small amount of energy sustains the entire living world

(3) One can find unidirectional flow of energy from the sun to producers and then to consumers

(4) All of these

30) Match the column :-

	Column-I		**Column-II**
a	Fragmentation	i	Leads to accumulation of a dark coloured substance
b	Humification	ii	Water-soluble inorganic nutrients undergo in soil horizon
c	Catabolism	iii	Bacterial and fungal enzymes degrade detritus into simple organic and inorganic substance
d	Leaching	iv	Detritivores break down detritus in small pieces

(1) a-iii, b-i, c-iv, d-ii

(2) a-iv, b-i, c-ii, d-iii

(3) a-iv, b-iii, c-i, d-ii

(4) a-iv, b-i, c-iii, d-ii

31) Choose the incorrect match :-

Column-A	**Column-B**

(1)	Annual net primary production of biosphere	→	170 billion ton
(2)	Productivity of ocean	→	55 billion ton
(3)	Pyramid of energy	→	Productivity
(4)	Secondary productivity	→	Autotrophs

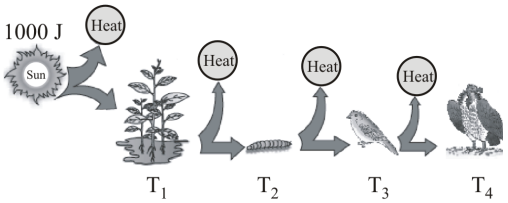
(1) 1

(2) 2

(3) 3

(4) 4

32) What is the magnitude of energy at top carnivores (T_4) level in given figure ?

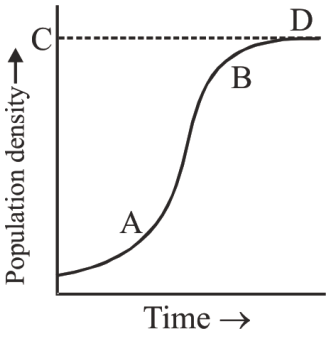


(1) 0.1 J

(2) 0.01 J

(3) 1 J

(4) 10 J



33) Which of the following is a correct match in the given sigmoid curve ?

	Column-I	**Column-II**
(i)	Carrying capacity	A
(ii)	Deceleration	B
(iii)	Asymptote	C
(iv)	Acceleration	D

Options :-

(1) (i)-B, (ii)-D, (iii)-C, (iv)-A

(2) (i)-B, (ii)-C, (iii)-D, (iv)-A

(3) (i)-D, (ii)-C, (iii)-A, (iv)-B

(4) (i)-C, (ii)-B, (iii)-D, (iv)-A

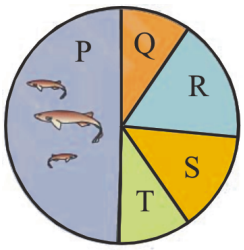
34) Population size more technically is designated as population density. Select correct statement with respect to this :-

(a) Not necessarily be measured in number only.
(b) Can be measured as biomass per unit area or volume.
(c) Number of individuals per unit area.

(1) Only (c) is correct
(2) Only (a) and (c) are correct
(3) Only (b) and (c) are correct
(4) (a), (b) and (c) are correct

35) In a month of January Siberian Cranes migrate from Russia to India for breeding. A survey was done -
· Till December total population of Siberian Cranes = 1200
· Birth Rate = 400
· Mortality Rate = 200
· Number of Cranes immigrated = 600
· Number of cranes emigrated = 300
Calculate the per capita Death rate :-

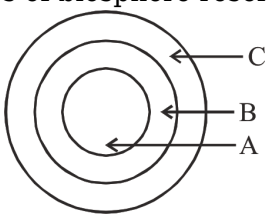
(1) 1.7×10^{-2}
(2) 200
(3) 17
(4) 1.7×10^{-1}

36) 

Above given pie chart is showing number of vertebrate species. In above pie chart identify S is representing species of :-

(1) Reptiles
(2) Amphibians
(3) Birds
(4) Fishes

37) Biosphere Reserve is a specified protected area in which multiple use of the land is permitted. In this three zones of biosphere reserve are found identify the correct zone (A, B, C) in given biosphere reserve zones.



	A	**B**	**C**
(1)	Transition zone	Buffer zone	Core zone

(2)	Buffer zone	Transition zone	Core zone
(3)	Core zone	Buffer zone	Transition zone
(4)	Core zone	Transition zone	Buffer zone

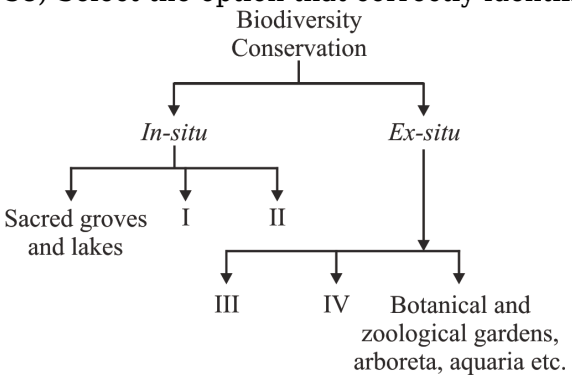
(1) 1

(2) 2

(3) 3

(4) 4

38) Select the option that correctly identifies I, II, III and IV :-



(1) I – Biosphere reserves
II – National parks, wildlife sanctuaries
III – Sacred plants at home gardens
IV – Gene banks, cryopreservation

(2) I – Sacred plants, home gardens
II – National parks, wildlife sanctuaries
III – Biosphere reserves
IV – Gene banks, cryopreservation

(3) I – Biosphere reserves
II – Gene banks, cryopreservation
III – Sacred plants, home gardens
IV – National parks, wildlife sanctuaries

(4) I – Biosphere reserves
II – Sacred plants, home gardens
III – National parks, wildlife sanctuaries
IV – Gene banks, cryopreservation

39) There are four major causes (the evil quartet) of biodiversity losses, with respect to them, which of the following statement is not correct ?

(1) Habitat loss and fragmentation decreases biodiversity

(2) Over exploitation cause some fishes to get endangered

(3) Alien species invasions decrease the threat to native species.

(4) Co-extinctions lead to extinction of a species as well as associated plant and animal species.

40) **Statement-I :-** At the species level Western Ghats have a greater amphibians species diversity than Eastern Ghats.

Statement-II :- The World summit an sustainable development held in 2002 in Johannesburg.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and Statement-II are incorrect
(3) Statement-I is incorrect and statement-II is correct
(4) Statement-I is correct and statement-II is incorrect

41) Given below are the two statements :
Statement-I :- Habitat loss and fragmentation is the most important cause driving animals and plants to extinction.
Statement-II :- Amazon forest is estimated to produce 20 percent of total oxygen in earth's atmosphere.
Select the correct option :

(1) Both statements are not correct
(2) Only statement-I is correct
(3) Only statement-II is correct
(4) Both statements are correct

42) **Assertion :** Biologists are not sure about how many prokaryotic species are present on earth.
Reason : Conventional taxonomic methods are not suitable for identification of microbial species.

(1) Both Assertion & Reason are True & Reason is the correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not the correct explanation of the Assertion.
(3) Assertion is True but Reason is False.
(4) Both Assertion & Reason are False.

43) **Assertion (A) :** The Abingdon tortoise in Galapagos Island became extinct within a decade.
Reason (R) : Tortoise became extinct due to the greater browsing efficiency of the goals.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

44) Choose the correct match for the following :-

	Column-I		**Column-II**
a.	Jhoom cultivation	i.	8.1% of the world
b.	Sacred grove	ii.	In-situ conservation
c.	Land area of India	iii.	2.4% of the world
d.	Biodiversity of India	iv.	Deforestation

(1) a-iv, b-ii, c-i, d-iii
(2) a-ii, b-iv, c-iii, d-i
(3) a-ii, b-iv, c-i, d-iii

(4) a-iv, b-ii, c-iii, d-i

45) Match the animal given in column-I with their place in column-II :-

Column-I		Column-II	
(i)	Dodo	(a)	Africa
(ii)	Quagga	(b)	Russia
(iii)	Thylacine	(c)	Mauritius
(iv)	Steller's sea cow	(d)	Australia

(1) i-a, ii-c, iii-b, iv-d

(2) i-d, ii-c, iii-a, iv-b

(3) i-c, ii-a, iii-b, iv-d

(4) i-c, ii-a, iii-d, iv-b

46) The table below gives the populations (in thousands) of ten species (A-J) in four areas (P-S) consisting of the number of habitats given within brackets against each. Study the table and answer the question which follows :

Area and Number of Habitats	Species and their populations (in thousands) in the areas									
	A	B	C	D	E	F	G	H	I	J
P(11)	10.2	0.62	1.53	8.2	-	6	1.1	11.2	9	-
Q(11)	-	0.52	2.3	12.3	7	1.5	8	8.3	1.1	10.3
R(13)	10.4	1.2	3.4	-	-	0.8	4.2	8.2	2.2	4.1
S(12)	0.4	4.6	1.5	7.1	9.3	0.8	3.3	7.3	11.3	2.1

Which area out of **P** to **S** shows maximum species diversity ?

(1) P

(2) Q

(3) R

(4) S

47)

Identify the correct match :-

Column – I	Column – II
(i) Species diversity	(a) Great influence on community stability
(ii) Species dominance	(b) Zonation according to the need of light
(iii) Stratification	(c) Different types of species in a community

(iv) Keystone species	(d) Highest no. of one type of species

(1) i–c, ii–b, iii–a, iv–d

(2) i–c, ii–d, iii–b, iv–a

(3) i–b, ii–a, iii–d, iv–c

(4) i–b, ii–d, iii–a, iv–c

48) Given below are two statements :
Statement-I : Ecologists believed that communities with more species, generally, tend to be more stable than those with less species.
Statement-II : A stable community should show too much variation in productivity from year to year; it must be resistant and resilient to occasional disturbances (natural or man-made), and it must also be resistant to invasions by alien species.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I is incorrect but Statement-II is correct

(2) Both Statement-I and Statement-II are correct

(3) Both Statement-I and Statement-II are incorrect

(4) Statement-I is correct but Statement-II is incorrect

49) Given below are two statements :
Statement-I : The diversity of plants and animals is not uniform throughout the world but shows a rather uneven distribution. For many group of animals or plants, there are interesting patterns in diversity, the most well-known being the latitudinal gradient in diversity, In general, species diversity increases as we move away from the poles towards the equator.
Statement-II : Colombia located near the equator has nearly 1,300 species of birds while New York at 41°N has 150 species and Greenland at 71°N only 46 species. India, with much of its land area in the tropical latitudes, has more than 200 species of birds.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I is incorrect but Statement-II is correct

(2) Both Statement-I and Statement-II are correct

(3) Both Statement-I and Statement-II are incorrect

(4) Statement-I is correct but Statement-II is incorrect

50) Identify the correct set of statements with regard to properties of humus.
(a) Completely resistant to microbial action.
(b) Dark coloured amorphous substance.
(c) End product of detritus food chain.
(d) Reservoir of nutrients.
(e) Undergoes decomposition very slow.
Choose the correct answer from the options given below :

(1) (a), (b) and (e) only

(2) (b) and (c) only

(3) (a), (b), (c), (d) and (e)

(4) (b), (d) and (e) only

51) Identify the correct match from the columns I, II and III.

	Column-I		Column-II		Column-III
p	Farmers friend	a	55 billion ton	i	In grassland
q	Productivity	b	detritivor	ii	bacteria
r	Pyramid of number	c	upright	iii	earthworm
s	Pyramid of biomass	d	Inverted	iv	In lake
		e	155 billion ton	v	In ocean

(1) p-b-ii, q-a-i, r-d-v, s-d-iv

(2) p-b-iii, q-e-v, r-d-iv, s-d-iv

(3) p-b-iv, q-a-i, r-c-iv, s-b-iii

(4) p-b-iii, q-a-v, r-c-i, s-d-v

52) Find the correct order of species diversity

(1) Insects > birds > fungi > fishes

(2) Mammals < amphibians < reptiles < bird

(3) Angiosperm > fishes > amphibians > reptiles

(4) Birds < mammals < reptiles < fishes

53) In Robert Paine's classic experiment, starfish (*Pisaster ochraceus*) were removed from an intertidal rocky shore. Within a year, mussels dominated the area, and species diversity of other invertebrates decreased drastically.
Which of the following statements are correct?
(a) Starfish acts as a keystone predator in the community.
(b) Removal of predator led to competitive exclusion among prey species.
(c) Predator presence helps maintain species diversity.
(d) Starfish removal decreases constrain of competition in invertebrate species.
Options :

(1) a, b, c, d only

(2) a and c only

(3) b, c, and d only

(4) a, b and c only

54) (a) Primary productivity depends on the plant species inhabiting a particular area.
(b) Detritus is the raw material for decomposition. The important steps in the process of decomposition are fragmentation, leaching, catabolism, humification and mineralisation.
(c) Secondary productivity also depends on a variety of environmental factors, availability of nutrients and photosynthetic capacity of plants.
(d) Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus,

(e) All animals depend on plants (directly or indirectly) for their food needs. They are hence called producer and also autotrophs
Choose the correct answer from the options given below :

(1) (a), (b) and (d) only

(2) (b) and (c) only

(3) (a), (b), (c), (d) and (e)

(4) (b), (d) and (e) only

55) Given below are two statements :
Statement-I : The number of trophic levels in the grazing food chain is restricted as the transfer of energy follows 10 per cent law – only 10 per cent of the energy is transferred to each trophic level from the higher trophic level.
Statement-II : An ecosystem can be visualised as a functional unit of nature, where living organisms interact among themselves and also with the surrounding physical environment.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) **Statement-I** is incorrect but **Statement- II** is correct

(2) Both **Statement-I** and **Statement-II** are correct

(3) Both **Statement-I** and **Statement-II** are incorrect

(4) **Statement-I** is correct but **Statement-II** is incorrect

56) (a) In nature, we can not find isolated, single individuals of any species
(b) Natural selection operates at the population level to evolve the desired traits
(c) A population at any given time is composed of individuals of different ages.
(d) Total number in population is not an easily adoptable measure if the population is huge and counting is impossible or very time-consuming.
(e) Natality refers to the number of births during a given period in the population that are added to the initial.
Choose the correct answer from the options given below :

(1) (a), (b) and (e) only

(2) (b) and (c) only

(3) (a), (b), (c), (d) and (e)

(4) (b), (d) and (e) only

57) More than ……**A**….. per cent of all the species recorded are animals, while plants (including algae, fungi, bryophytes, gymnosperms and angiosperms) comprise no more than ……**B**….. per cent of the total. Among animals, insects are the most species-rich taxonomic group, making up more than ……**C**….. A per cent of the total.

	A	**B**	**C**
1	70	21	70
2	60	22	77
3	77	23	60

4	70	22	70

(1) 1
(2) 2
(3) 3
(4) 4

58) Which is the effect of reduced species diversity in tropical rainforest?

(1) Decreased plant productivity
(2) Increased food web complexity
(3) Resistance for alien species enhanced
(4) Increased resilience to disturbance

59) genetic diversity is seen in

(1) Only in rice
(2) Only in wheat
(3) Only in mango
(4) In all livings

60) Initially …**A**……. biodiversity hotspots were identified but subsequently …**B**….. more have been added to the list, bringing the total number of biodiversity hotspots in the world to ……**C**

	A	**B**	**C**
1	25	11	36
2	29	7	36
3	24	10	34
4	25	9	34

(1) 1
(2) 2
(3) 3
(4) 4

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ANSWER KEYS

BIOLOGY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**2**	**2**	**2**	**1**	**1**	**2**	**2**	**2**	**2**	**1**	**4**	**1**	**3**	**1**	**2**	**2**	**4**	**3**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**4**	**4**	**2**	**2**	**4**	**3**	**4**	**4**	**4**	**4**	**2**	**4**	**4**	**4**	**1**	**3**	**1**	**3**	**1**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**4**	**1**	**1**	**4**	**4**	**4**	**2**	**2**	**4**	**4**	**4**	**2**	**4**	**1**	**1**	**3**	**4**	**1**	**4**	**4**

SOLUTIONS

BIOLOGY

1)

Bio Module # 8

2)

Bio Module # 8

3) **Question Explanation:**
The basic unit of ecological study is :-
Solution:
The basic unit of ecological study is the organism. It is the fundamental biological unit in ecology, representing an individual living being that interacts with its environment and other organisms. From the study of individual organisms, ecologists can move to higher levels of organization, such as populatins, communities, ecosystems, and the biosphere.
Hence,
Correct Option : 2

4)

NCERT XII # 205, 206

5) Module Pg. No. 05

6) Module Pg. No :- 05

7)

In most light receiving zone i.e. euphotic zone.

8) The relationship between the fig tree and the fig wasp is an example of mutualism, where both species benefit from the interaction. The fig tree provides a safe place for the fig wasp to lay its eggs inside its flowers. In return, the wasp pollinates the fig tree as it moves between flowers, ensuring that the tree reproduces. This mutualistic relationship is crucial for the fig tree's reproduction and the wasp's lifecycle.

9) XII NCERT Page No. # 201

10)

Question: What type of interaction shows in this graph between two organisms X and Y?
Correct Answer: 2. CommensalismExplanation:
In commensalism, one species benefits, while the other is unaffected. If the graph shows one species (say, X) growing or benefiting, while the other species (Y) shows no significant change, this would be a classic example of commensalism.

11)

NCERT XII Page # 201

12)

NCERT XII, Pg. # 197 to 200

13)

NCERT XII Page # 197

14) NCERT XII, Page # 234

15)

NCERT XII Page # 199, 197, 206

16) NCERT XII Pg.# 244

17)

The variation in Temperature shows the major effect on the rate of metabolism in organisms. Temperature plays a critical role in metabolic processes. It affects enzyme activity, the speed of chemical reactions, and overall cellular functions. In most organisms, metabolism increases with higher temperatures up to a certain point, after which it may decrease due to enzyme denaturation or other factors.

A. Rainfall, Light, and Water can influence metabolism, but Temperature has the most direct and profound effect on metabolic rates.

Therefore, the correct answer is Temperature.

18)

NCERT XII, Page # 209

19)

NCERT XII, Page No. 207

20) NCERT Pg. # 242

21) **Question Explanation :** What is stratification?
Concept : Ecosystem.
Solution : Stratification is vertical distribution of different species occupying different levels.
Final Answer : Option (3)

22)

Garden is the odd one out.
Forest, grassland, and desert are all natural ecosystems. A garden is a man-made or cultivated environment.

23)

The correct answer is: **Option 4: a, b, c, d**
Explanation: A **habitat** refers to the place where an organism lives and includes all the **biotic** (living) and **abiotic** (non-living) components that affect the organism. This includes:
• **Biotic components** (a): Other living organisms, such as plants, animals, and microorganisms.
• **Physio-chemical components** (b): Non-living elements like temperature, water, and soil type that influence the organisms.
• **Pathogens, parasites, predators, competitors** (c): These are living factors that affect survival and interactions within a habitat.
• **Different species (d):** The variety of species found in a particular habitat.
Thus, habitat is composed of all these factors (a, b, c, and d).

24)

Allen Module page no. 21

25)

Allen Module page no. 21

26) NCERT XII, Pg. # 211

27)

The correct answer is **3. (a) deer (b) rabbit (C) frog (d) rat**
Explanation: The given diagram shows representation of Food Web
FOOD WEB:

A. In a big ecosystem many food chains are interlinked together on different trophic levels to form a food web. In the food web, transfer of food energy is unidirectional but from many different alternative pathways.
B. In food web members of a particular trophic level obtain their food according to their choice

and taste but that type of facility is not present in the food chain. It means they have more than one option or alternative for getting food.

28)

NCERT (XII) Pg. # 249

29) NCERT XII, Pg. # 245

30) **Solution:**
The correct answer is 4. a-iv, b-i, c-iii, d-ii,
Explanation:
Let's break down each term

A. **Fragmentation (a):** This refers to the breaking down of detritus into smaller pieces by detritivores. This process is iv: Detritivores break down detritus into small pieces.
B. **Humification (b):** Humification is the process that leads to the accumulation of a dark-colored substance called humus. This process is i: Leads to accumulation of a dark-colored substance.
C. **Catabolism (c):** Catabolism involves the breakdown of complex organic matter into simpler substances. In this case, bacterial and fungal enzymes degrade detritus into simple organic and inorganic substances, so this matches iii: Bacterial and fungal enzymes degrade detritus into simple organic and inorganic substances.
D. **Leaching (d):** Leaching is the process by which water-soluble inorganic nutrients are removed from the soil horizon. This matches ii: Water-soluble inorganic nutrients undergo in soil horizon.

Final Matching:

A. a - iv: Fragmentation Detritivores break down detritus into small pieces
B. b - i: Humification Leads to accumulation of a dark-colored substance
C. c - iii: Catabolism inorganic substances Bacterial and fungal enzymes degrade detritus into simple organic and
D. d - ii: Leaching Water-soluble inorganic nutrients undergo in soil horizon

Thus, the correct answer is 4. a-iv, b-i, c-iii, d-ii.

31)

NCERT XII Page # 207, 212, 213

32)

NCERT XII, Page # 211

33)

NCERT Pg. # 194, 195

34)

NCERT-XII Pg.# 192

35)

NCERT-XII Pg#193

36)

NCERT XII Page # 218

37)

Bio Module # 8

38)

NCERT XII # 224, 225

39)

NCERT XII # 222

40) NCERT XII, Pg # 217,228

41) NCERT XII, Pg # 222,224

42) NCERT-XII Pg. # 218

43) NCERT XII, Page No. 234

44) NCERT (XII) Pg. # 261, 267, 284

45)

Explain Question : The question asks to correctly match extinct animals with the regions they were once found in.

Concept : This question is based on Cause of biodiversity loss.

Solution : Match each species to its correct region:
(i) Dodo → (c) Mauritius
(ii) Quagga → (a) Africa
(iii) Thylacine → (d) Australia

(iv) Steller's sea cow → (b) Russia

Final Answer : option (4)

46)

Calculate species diversity per habitat.

47)

Species diversity (i) corresponds to (c) Different types of species in a community.

A. **Explanation:** Species diversity refers to the variety of species within a community.

Species dominance (ii) corresponds to (d) Highest number of one type of species.

A. **Explanation:** Species dominance refers to the prevalence of one species that dominates the community structure.

Stratification (iii) corresponds to (b) Zonation according to the need of light.

A. **Explanation:** Stratification refers to the vertical layering in ecosystems based on light and other factors.

Keystone species (iv) corresponds to (a) Great influence on community stability.

A. **Explanation:** Keystone species play a critical role in maintaining the balance and stability of an ecosystem.

48)

NCERT XII # 221

49)

NCERT XII # 219

50)

NCERT XII # 207, 208

51)

NCERT XII # 207, 212, 213

52)

NCERT XII # 218

53)

NCERT XII # 197, 198

54)

NCERT XII # 207, 208

55)

NCERT XII # 206, 211

56)

NCERT XII # 191,192,193

57)

NCERT XII # 218

58)

NCERT XII # 220, 221

59)

NCERT XII # 217

60)

NCERT XII # 224